

Institute of Actuaries of India

Subject CT4 – Models

October 2015 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i) Benefits**

- Systems with long time frames can be studied in compressed time
- Complex systems with stochastic elements can be studied (especially by simulation modelling).
- Different future policies or possible actions can be compared either to see which best suits the requirements of a user or to examine different scenarios without carrying them out in practice, or to avoid potential costs associated with trying out in real life.
- Models allow control over experimental conditions, so that we can reduce the variance of the results output without upsetting the mean values.
- Parameters can be sensitivity tested using a model.

Limitations

- Model development requires a lot of time and expertise, and hence can be costly.
- May need to run model lots of times (essential if it is a stochastic model).
- Models are more useful for comparing the results of input variations than for optimising outputs.
- Models can look impressive, but can lull the user into a false sense of security. Impressive output is not a substitute for validity and close imitation of the real world.
- This is more true the further into the future you project
- Models rely heavily on the data input. If this is poor or lacking in credibility the output is likely to be flawed.
- Users need to understand the model sufficiently well to be able to know when it is appropriate to apply it.
- Interpretation of models can be difficult, and often outputs need to be seen in relative rather than absolute terms.
- Models cannot take into account all possible future events (e.g. changes in legislation)

[4]

ii) In assessing the suitability of a model for this particular exercise it is important to consider the following:

- the objectives of the modelling exercise, to recommend the number and capacity of colleges and hospitals required in this new City
- The simplicity of the model; that the model is not overly complex so that the model is relatively straightforward to explain to the planners/developers
- Should consider whether there are trends in birth rates, rather than simply using current rates.
- The data specified are likely to be available from reliable sources and the data should be validated against alternate sources
- Although it may be possible that the starting point for the planned population is not appropriate
- Unforeseen events may take place such as a national epidemic which changes the rates.

- Current age distribution for the area may not be representative of that for the new Smart city as, for example, semi urban areas may have different distributions to urban areas
- May wish to consider experience of similar type of smart city for data credibility.
- Migration may affect the profile of the population, for example older families moving away and younger families buying their houses may mean the age structure remains relatively constant over time regardless of mortality and fertility rates.
- the possible errors associated with the model or parameters used not being a perfect fit, representation of the underlying situation being modelled,
- the impact of correlations between the random variables that drive the model, whether the same has been appropriately captured in the model development
- the relevance of model given the type of models used for similar projection while planning for other smart cities

[4]

iii) Points need to consider during design of a model used to project over only a short time frame may differ from one used to project over long years:

- A variable which has an ignorable effect in the short term may have a non-ignorable effect over long term.
- Over the short term, it may be reasonable to assume the values of some variables to be constant or to vary linearly, whereas this would not be reasonable over long term.
- For example, growth which is exponential may appear linear if studied over a short time frame.
- The interaction between variables in the short-term may be different from that over the long term.
- Higher order relationships between variables may be ignored for simplicity if modelling over a short time frame.
- The time units used in the model might be shorter for a model projecting over a short time frame, so that the total number of time units used in each model is roughly the same.
- Over long years, regulatory changes and other “shock” events are more likely to occur, and the model design may need to consider the circumstances in which the results or conclusions may be materially impacted
- e.g. in the short term the tax basis may be known, but in the long run it is likely to change).

[4]

[12 Marks]

Solution 2:

From the lives which survived till end of year, it can be seen that $q_{70} = 4,350/1,00,000 = 0.0435$, $p_{70} = 0.9565$

i) Under uniform distribution of deaths $1-t q_{x+t} = t.q_x$

Hence $0.5 p_{70} = 1- 0.5 q_{70} = 1- 0.5 * q_{70} = 1 - 0.5*0.0435= 1- 0.02175 = 0.97825$

$0.5 p_{70.5} = 1 p_{70} / 0.5 p_{70} = 0.9565/ 0.97825 = 0.97777$

[1]

ii) Under Balducci assumption

$1-t q_{x+t} = (1-t)*q_x$

Hence $0.5 p_{70.5} = 1- 0.5 q_{70} = 1- (1-0.5)*0. 0.0435= 1 - 0.02175= 0.97825$

$0.5 p_{70} = 1 p_{70} / 0.5 p_{70.5} = 0.9565/ 0.97825 = 0.97777$

[1]

iii) Under constant force of mortality

$$tq_x = 1 - e^{-\mu t}, \text{ where } \mu = -\ln(1 - q_{70}) = 0.044474$$

$$0.5 p_{70} = e^{-\mu t} = 0.978008$$

$$0.5 p_{70.5} = 0.978008$$

[1]

[3 Marks]

Solution 3:

i) A Markov chain is a stochastic process with discrete states operating in discrete time in which the probabilities of moving from one state to another are dependent only on the present state of the process.

The chain is said to be time-homogeneous Markov Chain if either of the conditions is met

- the transition probabilities are also independent of time.
- the 1-step transition probabilities are dependent only on the time lag, the chain is said to be time-homogeneous.

[2]

ii) a) A Markov chain with a finite state space has at least one stationary probability distribution. [1]

b) An irreducible Markov chain with a finite state space has a unique stationary probability distribution. [1]

c) A Markov chain with a finite state space which is irreducible, and which is also aperiodic converges to a unique stationary probability distribution. [1]

[5 Marks]

Solution 4:

i) Rate interval is defined as the period during which the age remains constant. The age classification of the lapsing data is age last birthday on the policy anniversary prior to lapsing. As the age remains constant during the policy year, it is a policy-year rate interval. [1]

ii) The estimate of the lapse rate would be applicable to the age at the middle of the rate interval.

Assuming birthdays are uniformly distributed over the policy year, the average age at the start of the interval is $x+1/2$ and at the middle of the interval is $x+1$. Hence the age for which the estimated lapse rate for x is applicable, is $x+1$. [2]

iii) As the age at lapse is defined as age x last birthday at the preceding policy anniversary, the corresponding exposed to risk should be defined similarly.

Let $P_{x,t}$ be the number of policies in force aged x last birthday at the preceding policy anniversary at time t .

Then, if t is measured in years since 1 January 2011, a consistent exposed-to risk would be

$$E_c^x = \int_0^1 P_{x,t}$$

which, assuming that policy anniversaries are uniformly distributed across the calendar year, may be approximated

as

$$E_c^x = \frac{1}{2} [P_{x,0} + P_{x,1}] \quad [2]$$

iv) The data given is the number of policies in force at time t , classified by age last birthday at time $t(P'_{x,t})$.

A life aged x last birthday on the policy anniversary prior to lapsing would be aged ' x ' at time t if no birthday falls between the policy anniversary and time ' t ' and would be aged ' $x+1$ '

If a birthday has fallen between the policy anniversary and time ' t '.

Assuming that birthdays are uniformly distributed across the policy year, half of these lives will be aged x last birthday and half will be aged $x+1$ last birthday.

$$P_{x,t} = \frac{1}{2} [P'_{x,t} + P'_{x+1,t}]$$

Hence the required exposed to risk is

$$E_c^x = \frac{1}{2} \left[\frac{1}{2} [P'_{x,0} + P'_{x+1,0}] + \frac{1}{2} [P'_{x,1} + P'_{x+1,1}] \right] \quad [2]$$

[7 Marks]

Solution 5:

i) Consider a small time interval dt . The probability of an arrival of claim from the individual product in time dt is $\lambda \cdot dt + o(dt)$ and

the probability of a arrival from the Group Scheme in time dt is $\mu \cdot dt + o(dt)$.

The arrival probability for the sum of the claims in dt is therefore

$$(\lambda + \mu) \cdot dt + o(dt)$$

This is by definition a Poisson process with rate $(\lambda + \mu)$. [3]

ii) A jump chain is formed by recording the state of a Markov jump process only at the instant when a transition has just been made. The jump chain is in itself a Markov chain. [1]

iii) The outcome of the jump chain can only differ from that of the standard Markov chain if the jump process enters an absorbing state.

As the jump process will make no further transitions once it enters an absorbing state, the jump chain "stops".

It is possible to model the jump chain as though transitions continue to occur but the chain continues to occupy the same state. [2]

[6 Marks]

Solution 6:

Consider the durations t_j at which events take place.

Let the number of deaths at duration t_j be d_j and the number of pensioners still at risk of death at duration t_j be n_j .

At $t_j = 1$, $\lambda(t)$ is 0.05.

Since the Nelson Aalen estimate of $\lambda(t)$ is $\sum_{j=1}^{j=t} d_j / n_j$

We have $\lambda(1) = d_1/n_1 = 0.05$.

As all pensioners are at risk of dying at t_1 , $d_1/20 = 0.05$

$$d_1=1.$$

At $t=2$, we have $d_2/n_2 = 0.175 - 0.05 = 0.125$. We can have at most 19 pensioners at risk and since deaths would be in integers $d_2 = 2$, $n_2 = 16$. The number of pensioners transferring the account (c_1) is 3.

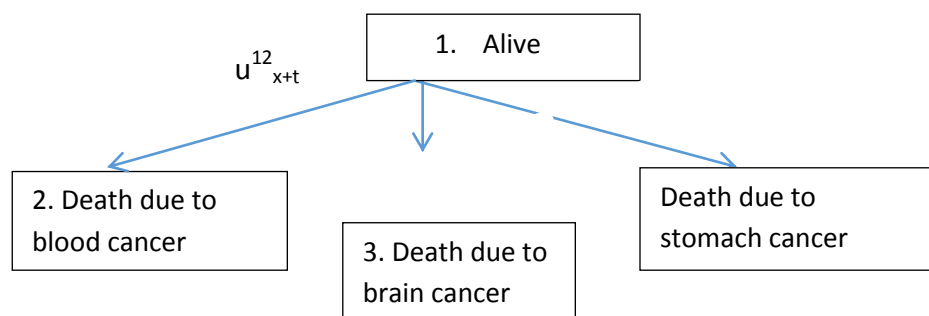
The following table can be derived following similar logic

t_j	n_j	d_j	c_j	
1	20	1	3	
2	16	2	2	
3	12	3	0	
4	9	3	1	
5	5	1	4	

[3+3=6 Marks]

Solution 7:

i)



[2]

ii) Assuming the process follows a Markov Process,

we have

$$t+dt P^x_{12} = t P^x_{11} * dt P^x_{12} + t p^x_{12} * dt P^x_{22}$$

since it is impossible to leave states 3 and 4 once entered.

And also, ${}_tP_{x+t}^{22} = 1$,

since state 2 is an absorbing state.

We now assume that, for small dt ,

$$dt P_{x+t}^{12} = \mu_{x+t}^{12} dt + o(dt)$$

where $o(dt)$ is the probability that a life makes two or more transitions in the time interval dt , and

$$\lim_{dt \rightarrow 0} o(dt)/dt = 0$$

Substituting for $dt P_{x+t}^{12}$ gives

$${}_tP_{x+dt}^{12} = \mu_{x+t}^{12} \cdot {}_tP_{x+t}^{11} dt + {}_tP_{x+t}^{12} + o(dt)$$

Thus

$${}_tP_{x+dt}^{12} - {}_tP_x^{12} = \mu_{x+t}^{12} \cdot {}_tP_x^{11} dt + o(dt)$$

and

$$d/dt ({}_tP_x^{12}) = \lim_{dt \rightarrow 0+} ({}_tP_{x+dt}^{12} - {}_tP_x^{12})/dt = \mu_{x+t}^{12} \cdot {}_tP_x^{11}$$

[5]

[7marks]

Solution 8:

- i) The Poisson model is used for modelling the number of events that occur during a given period of time. Hence it can also be used to model the number of deaths among a group of lives, given the time spent exposed to risk.

[1]

- ii) The Poisson model is not an exact model, since it allows for a non-zero probability of more than n deaths in a sample of size n .

[1]

- iii) The contributions to the Poisson likelihood made by each member are proportional to the following

Ref No	Date of birth	Date of taking policy	Status as on 31 st March 2014	Date of exit (wherever applicable)	50th birthday	51st birthday	Start date latest of (50th birthday, Start of investigation, policy issue)	End date (earliest of 51st birthday, exit, end of investigation)	Duration exposed	
A	1 st April 1963	10 th Jan 2011	In-force	-	1st April 2013	1st April 2014	1st April 2013	31st March 2014	12 months	$\exp(-1*\mu_{50})$
B	1 st October 1963	1 st May 2013	In-force	-	1 st October 2013	1 st October 2014	1 st October 2013	31st March 2014	6 months	$\exp(-0.5*\mu_{50})$
C	1 st November 1962	25 th March 2012	Died	1 st December 2013	1 st November 2012	1 st November 2013	1st April 2013	1 st November 2013	7 months	$\exp(-7/12*\mu_{50})$
D	1 st May 1962	20 th July 2010	Died	1 st July 2013	1 st May 2012	1 st May 2013	1st April 2013	1 st May 2013	1 month	$\exp(-1/12*\mu_{50})$
E	1 st July 1964	1 st June 2000	Lapsed	1 st June 2013	1 st July 2014	1 st July 2015	1 st July 2014	31st March 2014	-	-
F	1 st March 1963	1 st May 2000	In-force		1st March 2013	1st March 2014	1st April 2013	1st March 2014	11 months	$\exp(-11/12*\mu_{50})$
G	1 st January 1963	1 st April 2009	Lapsed	1 st April 2013	1st January 2013	1st January 2014	1st April 2013	1 st April 2013	-	-
H	1 st July 1962	1 st Aug 2013	In-force		1st July 2012	1st July 2013	1 st Aug 2013	1st July 2013	-	-
I	1 st Feb 1963	1 st Feb 2013	In-force		1st Feb 2013	1st Feb 2014	1st April 2013	1st Feb 2014	10 months	$\exp(-10/12*\mu_{50})$
J	1 st May 1963	15 th Dec 2008	Died	1 st March 2014	1st May 2013	1st May 2014	1st May 2013	1 st March 2014	10 months	$\mu_{50} * \exp(-10/12*\mu_{50})$

The total likelihood, L , is proportional to the product

$$[\exp(-4.75 * \mu_{50})] * (\mu_{50})$$

Taking log

$$\log L = -4.75 * \mu_{50} + \log \mu_{50}$$

$$\frac{d \log L}{d \mu_{50}} = -4.75 + \frac{1}{\mu_{50}}$$

Setting this equal to zero and solving for μ_{50} produces the maximum likelihood estimate,

which is 0.2105

Since $\frac{d^2 \log L}{d \mu_{50}^2} = -\frac{1}{\mu_{50}^2}$, which is always negative, we have a maximum. [6]

iv) The Binomial model requires the lives to be observed for the entire period from age 50 to 51, which is not required under the Poisson model.

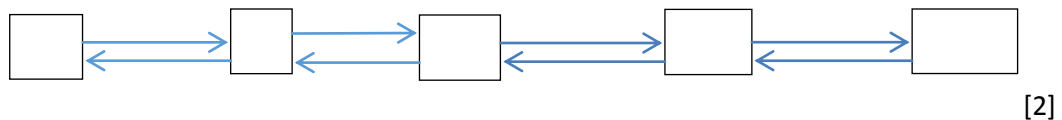
The Poisson model can be used in situations where there are decrements other than that being investigated, which is not possible in the Binomial model [2]

[10 Marks]

Solution 9:

i) The states can be labelled as 0, 1, 2:0, 2:1, 2:2, where 0 and 1 represent the number of employees occupied and 2:j means that both employees are occupied and j customers are waiting. [2]

ii)



iii) Transition Matrix (A) for the Markov jump chain is

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 2/5 & 0 & 3/5 & 0 & 0 \\ 0 & 4/7 & 0 & 3/7 & 0 \\ 0 & 0 & 4/7 & 0 & 3/7 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

The stationary probability distribution is

$$\pi A = \pi$$

$$1/2 \pi_0 = 1/3 \pi_1$$

$$\pi_1 = 3/2 \pi_0$$

$$5/6 \pi_1 = 1/2 \pi_0 + 2/3 \pi_{2:0}$$

$$\pi_{2:0} = 9/8 \pi_0$$

$$7/6 \pi_{2:0} = 1/2 \pi_1 + 2/3 \pi_{2:1}$$

$$\pi_{2:1} = 27/32 \pi_0$$

$$7/6 \pi_{2:1} = 1/2 \pi_{2:0} + 2/3 \pi_{2:2}$$

$$\pi_{2:2} = 81/128 \pi_0$$

To solve this we add the condition that the π_i must sum to 1.

$$\pi = (0.1960, 0.2940, 0.2205, 0.1654, 0.1240)$$

[5]

[9 Marks]

Solution 10:

i) Under the Cox model each individual's hazard is proportional to the baseline hazard, with the constant of proportionality depending on certain measurable quantities called co-variables. Hence the model is also called a proportional hazards model. [1]

ii) $\lambda(t) = \lambda_0(t) \exp(\beta_F * F + \beta_M * M + \beta_D * D)$, where $\lambda(t)$ is the estimated hazard and $\lambda_0(t)$ is the baseline hazard. [2]

iii) The baseline hazard refers to annual policy taken through the Online channel and where premiums are paid by direct debit [1]

iv) The results imply that

$$\exp[(\beta_D * 1)] / \exp[(\beta_D * 1) + \beta_F * 1 + \beta_M * 1] = 0.75 \quad \text{i.e}$$

$$\exp(\beta_F + \beta_M) = 4/3 \quad \text{Eqn 1}$$

$$\exp(\beta_D * 1) / \exp[(\beta_F * 1)] = 1 \quad \text{Eqn 2}$$

$$\exp(\beta_M * 1) / \exp[(\beta_D * 2)] = 0.75 \quad \text{Eqn 3}$$

Substituting from (2) into (1) gives

$$\exp(\beta_D + \beta_M) = 4/3$$

$$\exp(\beta_D) * \exp(\beta_M) = 4/3 \quad \text{Eqn 4}$$

From Eqn 3

$$(\exp(\beta_D))^2 * 0.75 = \exp(\beta_M)$$

So

Substituting in Eqn 4

$$\exp(\beta_D) * (\exp(\beta_D))^2 * 0.75 = 4/3$$

$$(\exp(\beta_D))^3 = 1.7778$$

$$\exp(\beta_D) = 1.2114$$

$$\beta_D = 0.19179$$

$$\beta_F = 0.19179$$

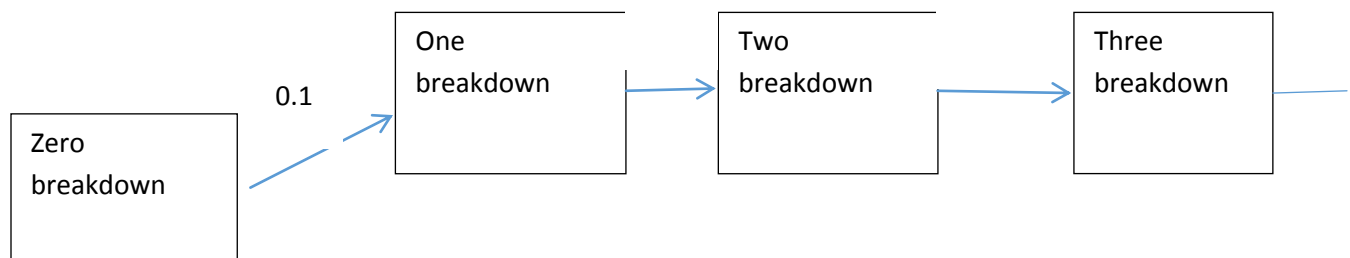
$$\beta_M = 0.0959$$

[6]

[10 Marks]

Solution 11:

i)



[1]

ii)

$$P'_0(t) = -0.1 * P_0(t)$$

$$P'_1(t) = 0.1 * P_0(t) - 0.2 * P_1(t)$$

$$P'_2(t) = 0.2 * P_1(t) - 0.25 * P_2(t)$$

[2]

iii)

a) Dividing the first equation by $P_0(t)$:

$$d/dt [\ln P_0(t)] = -0.1$$

Hence, using the boundary condition $P_0(0) = 1$

$$P_0(t) = \exp(-0.1*t)$$

[1]

b) Substitute into the second equation above to obtain

$$P'_1(t) = 0.1 * \exp(-0.1*t) - 0.2 * P_1(t)$$

$$\exp(0.2*t) * [P'_1(t) + 0.2 * P_1(t)] = 0.1 * \exp(-0.1*t + 0.2*t)$$

$$d/dt [\exp(0.2*t) * P_1(t)] = 0.1 * \exp(0.1*t)$$

$$\exp(0.2*t) * P_1(t) = \exp(0.1*t) + C$$

$$P_1(t) = \exp(-0.1*t) + C * \exp(-0.2*t)$$

$$P_1(0) = 0 \text{ as boundary condition}$$

$$P_1(t) = \exp(-0.1*t) - \exp(-0.2*t) \quad [2]$$

iv) In a similar way with the equation for $P_2(t)$

$$P'_2(t) = 0.2 * \exp(-0.1*t) - 0.2 * \exp(-0.2*t) - 0.25 * P_2(t)$$

$$d/dt [\exp(0.25*t) * P_2(t)] = 0.2 * [\exp(0.15*t) - \exp(0.05*t)]$$

$$\exp(0.25*t) * P_2(t) = 1.333 * \exp(0.15*t) - 4 * \exp(0.05*t) + 2.667$$

$$P_2(t) = 1.333 * \exp(-0.10*t) - 4 * \exp(-0.20*t) + 2.667 \exp(-0.25*t) \quad [3]$$

v) Expected claims = $1 * P_1(t) + 2 * P_2(t) + 3 * \sum_{i=3}^{\infty} P_i(1)$

$$= 1 * P_1(t) + 2 * P_2(t) + 3 * (1 - P_0(1) - P_1(1) - P_2(1))$$

Substitute the values

$$P_0(1) = 0.91$$

$$P_1(1) = 0.086$$

$$P_2(1) = 0.00833$$

$$\text{Expected claim} = 0.105 \quad [3]$$

[12 Marks]

Solution 12:

i) The crude mortality rates derived from the data will progress more or less roughly. However, it is expected that mortality rates would be smooth functions of age. Hence it follows that a crude estimate at any age 'x' also carries information about the mortality rates at age 'x-1' and 'x+1'.

Graduating the crude rates will enable this information to be captured, thus resulting in the mortality rates progressing smoothly over age.

The mortality rates would be used in computing premium rates/annuity rates. If the underlying mortality rates do not progress smoothly, then the derived premium rates etc would also fluctuate randomly, which would be difficult to understand/explain to potential customers.

The purpose of graduation is

- To produce a smooth set of rates that are suitable for a practical purpose
- To remove random sampling errors
- To use the information available from adjacent ages

[2]

ii) The methods used for graduation are

- Graduation by parametric formula
- Graduation by reference to a standard table
- Graphical graduation

[1]

iii) The null hypothesis is that the graduated rates represent the mortality of the members in the pension scheme.

To test the appropriateness of the graduation using chi-squared test, we compare $\sum_x z_x^2$ with χ^2 with 'm' degrees of freedom.

Age-group	Pensioners	Actual deaths	Exp deaths	Stddevn	zx2
50-54	37,259	248	227.28	1.3786	1.900553
55-59	28,057	392	367.55	1.2839	1.648501
60-64	25,654	680	672.13	0.3074	0.094513
65-69	20,475	987	997.13	-0.3290	0.108234
70-74	16,219	1380	1360.77	0.5445	0.296513
75-79	11,843	1625	1584.59	1.0906	1.189511
80-84	7,532	1564	1487.57	2.2121	4.893339
85-89	3,294	925	891.36	1.3195	1.740956
90+	450	130	155.48	-2.5254	6.377613

The number of age-groups is 9. As the graduation has been done with reference to a standard table, some degrees of freedom are lost.

So $m < 9$, say $m = 8$.

The value of $\sum_x z_x^2$ is 18.2497, which is higher than the critical value of chi-squared distribution with 8 degrees of freedom at 5 percent level (which is 15.51).

Hence we reject the null hypothesis.

[4]

iv) Any two from the below

Signs test

The signs test checks for any overall bias, whether the graduated rates are too high or too low.

The number of positive signs (or negative) is a Binomial distribution $B(9,0.5)$. There are 7 positive deviations and the probability of obtaining 7 or more positive signs is 0.01953 and since this is lower than 2.5%, we reject the null hypothesis and conclude that the graduated rates do not adhere to the data.

Individual Standardised deviations test

The Individual Standardised Deviations tests looks for individual large deviations at particular ages.

b. If the graduated rates were the true rates underlying the observed rates we would expect the individual deviations to be distributed Normal $(0,1)$ and therefore only 1 in 20 zxs should have absolute magnitudes greater than 1.96.

Looking at the zxs we see that there are 2 deviations higher than 1.96 and hence we reject the null hypothesis that the graduated rates are the true rates underlying the crude rates.

Cumulative deviations test

The cumulative deviations test detects overall bias or long runs of deviations of the same sign.

The test statistic

$$\sum (\phi_x - E_x q_x^o) / \sqrt{\sum E_x q_x^o} \text{ is } N(0,1)$$

$$= 187.14 / 87.99 = 2.1265$$

Since this value is higher than 1.96, the critical value at the 2.5% level, we reject the null hypothesis. [4]

v) The graduated rates are below the crude mortality rates for most of the ages, the exceptions being 65-69 and 90+ age group.

Since the graduated rates under-estimate the actual mortality experience, use of the graduated rates may result in over-provisioning of the annuity liabilities under the pension scheme. [2]

[13 Marks]
